Introduction to linear regression

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The Human Freedom Index is a report that attempts to summarize the idea of "freedom" through a bunch of different variables for many countries around the globe. It serves as a rough objective measure for the relationships between the different types of freedom - whether it's political, religious, economical or personal freedom - and other social and economic circumstances. The Human Freedom Index is an annually co-published report by the Cato Institute, the Fraser Institute, and the Liberales Institut at the Friedrich Naumann Foundation for Freedom.

In this lab, you'll be analyzing data from Human Freedom Index reports from 2008-2016. Your aim will be to summarize a few of the relationships within the data both graphically and numerically in order to find which variables can help tell a story about freedom.

Getting Started

Load packages

In this lab, you will explore and visualize the data using the **tidyverse** suite of packages. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let's load the packages.

```
library(tidyverse)
library(openintro)
data('hfi', package='openintro')
```

The data

The data we're working with is in the openintro package and it's called `hfi`, short for Human Freedom Index.

1. What are the dimensions of the dataset?

Insert your answer here

```
dim(hfi)
```

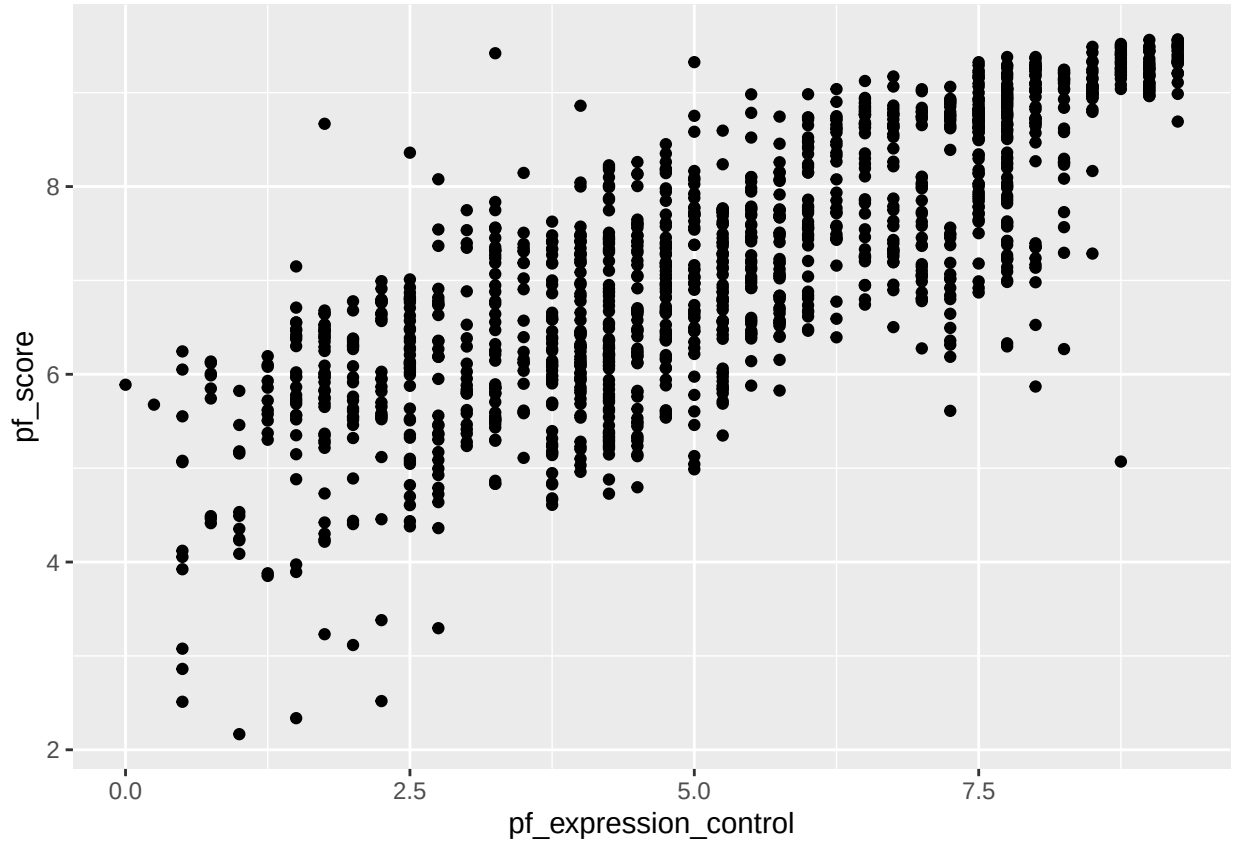
```
## [1] 1458  123
```

The dataset has 1,458 rows and 123 columns

2. What type of plot would you use to display the relationship between the personal freedom score, `pf_score`, and one of the other numerical variables? Plot this relationship using the variable `pf_expression_control` as the predictor. Does the relationship look linear? If you knew a country's `pf_expression_control`, or its score out of 10, with 0 being the most, of political pressures and controls on media content, would you be comfortable using a linear model to predict the personal freedom score?

Insert your answer here

```
ggplot(data = hfi, aes(x = pf_expression_control, y = pf_score)) +
  geom_point()
```



The relationship looks roughly linear as `pf_score` rises steadily along `pf_expression_control`. Since the pattern is linear, it would be reasonable to use a linear model to predict personal freedom from `pf_expression_control`, though predictions would still carry noticeable error

If the relationship looks linear, we can quantify the strength of the relationship with the correlation coefficient.

```
hfi %>%
  summarise(cor(pf_expression_control, pf_score, use = "complete.obs"))
```

```
## # A tibble: 1 x 1
##   `cor(pf_expression_control, pf_score, use = "complete.obs")`
##                                                          <dbl>
## 1                                                        0.796
```

Here, we set the `use` argument to "complete.obs" since there are some observations of NA.

Sum of squared residuals

In this section, you will use an interactive function to investigate what we mean by "sum of squared residuals". You will need to run this function in your console, not in your markdown document. Running the function also requires that the `hfi` dataset is loaded in your environment.

Think back to the way that we described the distribution of a single variable. Recall that we discussed characteristics such as center, spread, and shape. It's also useful to be able to describe the relationship of two numerical variables, such as `pf_expression_control` and `pf_score` above.

3. Looking at your plot from the previous exercise, describe the relationship between these two variables. Make sure to discuss the form, direction, and strength of the relationship as well as any unusual observations.

Insert your answer here

There is a positive correlation between the variables, which means that the lower the political pressure and control over the content of the media, the higher the level of personal freedom score will be r= 0.80. This relationship has an approximate linear form and is strong enough. Due to the fact that pf_expression_control uses discrete values, the dots are grouped into vertical strips, and there is a lot of overlap.

Just as you've used the mean and standard deviation to summarize a single variable, you can summarize the relationship between these two variables by finding the line that best follows their association. Use the following interactive function to select the line that you think does the best job of going through the cloud of points.

```
# This will only work interactively (i.e. will not show in the knitted document)
hfi <- hfi %>% filter(complete.cases(pf_expression_control, pf_score))
DATA606::plot_ss(x = hfi$pf_expression_control, y = hfi$pf_score)
```

After running this command, you'll be prompted to click two points on the plot to define a line. Once you've done that, the line you specified will be shown in black and the residuals in blue. Note that there are 30 residuals, one for each of the 30 observations. Recall that the residuals are the difference between the observed values and the values predicted by the line:

$$e_i = y_i - \hat{y}_i$$

The most common way to do linear regression is to select the line that minimizes the sum of squared residuals. To visualize the squared residuals, you can rerun the plot command and add the argument `showSquares = TRUE`.

```
DATA606::plot_ss(x = hfi$pf_expression_control, y = hfi$pf_score, showSquares = TRUE)
```

Note that the output from the `plot_ss` function provides you with the slope and intercept of your line as well as the sum of squares.

4. Using `plot_ss`, choose a line that does a good job of minimizing the sum of squares. Run the function several times. What was the smallest sum of squares that you got? How does it compare to your neighbors?

Insert your answer here

The smallest sum of squares I got was 953.

The linear model

It is rather cumbersome to try to get the correct least squares line, i.e. the line that minimizes the sum of squared residuals, through trial and error. Instead, you can use the `lm` function in R to fit the linear model (a.k.a. regression line).

```
m1 <- lm(pf_score ~ pf_expression_control, data = hfi)
```

The first argument in the function `lm` is a formula that takes the form `y ~ x`. Here it can be read that we want to make a linear model of `pf_score` as a function of `pf_expression_control`. The second argument specifies that R should look in the `hfi` data frame to find the two variables.

The output of `lm` is an object that contains all of the information we need about the linear model that was just fit. We can access this information using the summary function.

```
summary(m1)
```

```
##
## Call:
## lm(formula = pf_score ~ pf_expression_control, data = hfi)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.8467 -0.5704  0.1452  0.6066  3.2060
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            4.61707    0.05745   80.36   <2e-16 ***
## pf_expression_control  0.49143    0.01006   48.85   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8318 on 1376 degrees of freedom
##   (80 observations deleted due to missingness)
## Multiple R-squared:  0.6342, Adjusted R-squared:  0.634
## F-statistic:  2386 on 1 and 1376 DF,  p-value: < 2.2e-16
```

Let's consider this output piece by piece. First, the formula used to describe the model is shown at the top. After the formula you find the five-number summary of the residuals. The "Coefficients" table shown next is key; its first column displays the linear model's y-intercept and the coefficient of `pf_expression_control`. With this table, we can write down the least squares regression line for the linear model:

$$\hat{y} = 4.61707 + 0.49143 \times pf_expression_control$$

One last piece of information we will discuss from the summary output is the Multiple R-squared, or more simply, R^2. The R^2 value represents the proportion of variability in the response variable that is explained by the explanatory variable. For this model, 63.42% of the variability in `pf_free` is explained by `pf_expression_control`.

5. Fit a new model that uses `pf_expression_control` to predict `hf_score`, or the total human freedom score. Using the estimates from the R output, write the equation of the regression line. What does the slope tell us in the context of the relationship between human freedom and the amount of political pressure on media content?

Insert your answer here

```
m2 <- lm(hf_score ~ pf_expression_control, data = hfi)
summary(m2)
```

```
##
## Call:
## lm(formula = hf_score ~ pf_expression_control, data = hfi)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.6198 -0.4908  0.1031  0.4703  2.2933
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)           5.153687   0.046070  111.87   <2e-16 ***
## pf_expression_control 0.349862   0.008067   43.37   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.667 on 1376 degrees of freedom
##   (80 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.5775, Adjusted R-squared:  0.5772
## F-statistic:  1881 on 1 and 1376 DF,  p-value: < 2.2e-16
```

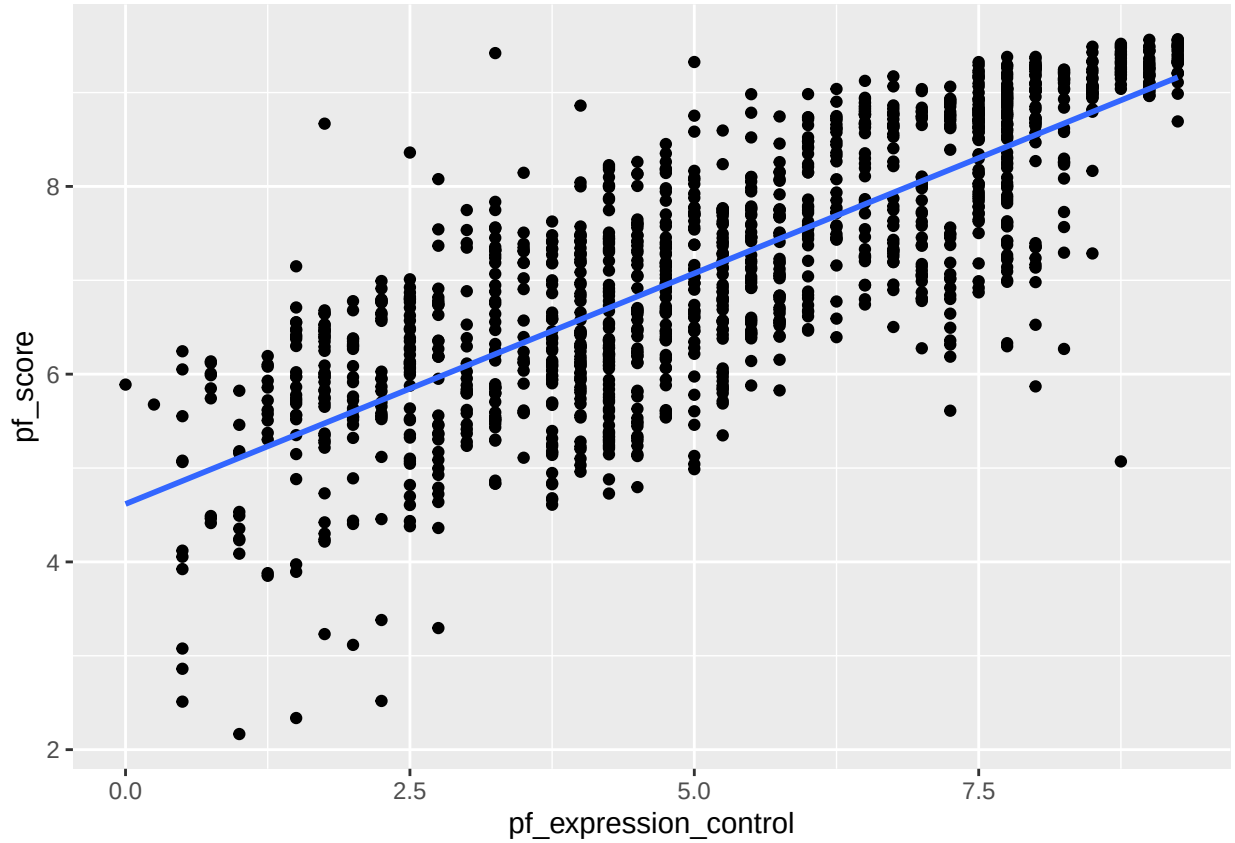
$$\hat{hf_score} = 5.15369 + 0.34986 \times pf_expression_control$$

This means that on average, for every unit change in pf_expression_control, there is a corresponding change of 0.35 units in the total human freedom score. The intercept is 5.15; this means that the total human freedom score of a country that has a value of zero in pf_expression_control is 5.15. Since $R^2 = 0.5775$, this means that 57.8% of the variability in human freedom score is explained by pf_expression_control.

Prediction and prediction errors

Let's create a scatterplot with the least squares line for `m1` laid on top.

```
ggplot(data = hfi, aes(x = pf_expression_control, y = pf_score)) +
  geom_point() +
  stat_smooth(method = "lm", se = FALSE)
```



Here, we are literally adding a layer on top of our plot. `geom_smooth` creates the line by fitting a linear model. It can also show us the standard error `se` associated with our line, but we'll suppress that for now.

This line can be used to predict y at any value of x. When predictions are made for values of x that are beyond the range of the observed data, it is referred to as *extrapolation* and is not usually recommended. However, predictions made within the range of the data are more reliable. They're also used to compute the residuals.

6. If someone saw the least squares regression line and not the actual data, how would they predict a country's personal freedom school for one with a 6.7 rating for `pf_expression_control`? Is this an

overestimate or an underestimate, and by how much? In other words, what is the residual for this prediction?

Insert your answer here

Based on the regression line: $\hat{y} = 4.61707 + 0.49143 \times 6.7 \approx 7.91$. Overestimate/underestimate is based on what the true country value is. There are no observations equal to 6.7 and countries at that point vary from around 6.5 (Myanmar, 2009) to 9.2 (Spain, 2010). For instance, Chile in 2016 has pf_expression_control = 6.75 and a true pf_score of approximately 8.22, and thus the regression line will underestimate it by about 8.22 - 7.94 = +0.28. France (2016, 8.77) will be a residual of approximately +0.83, whereas Belize (2016, 7.43) will be an overestimate with a residual of about -0.51.

```
predict(m1, data.frame(pf_expression_control = 6.7))
```

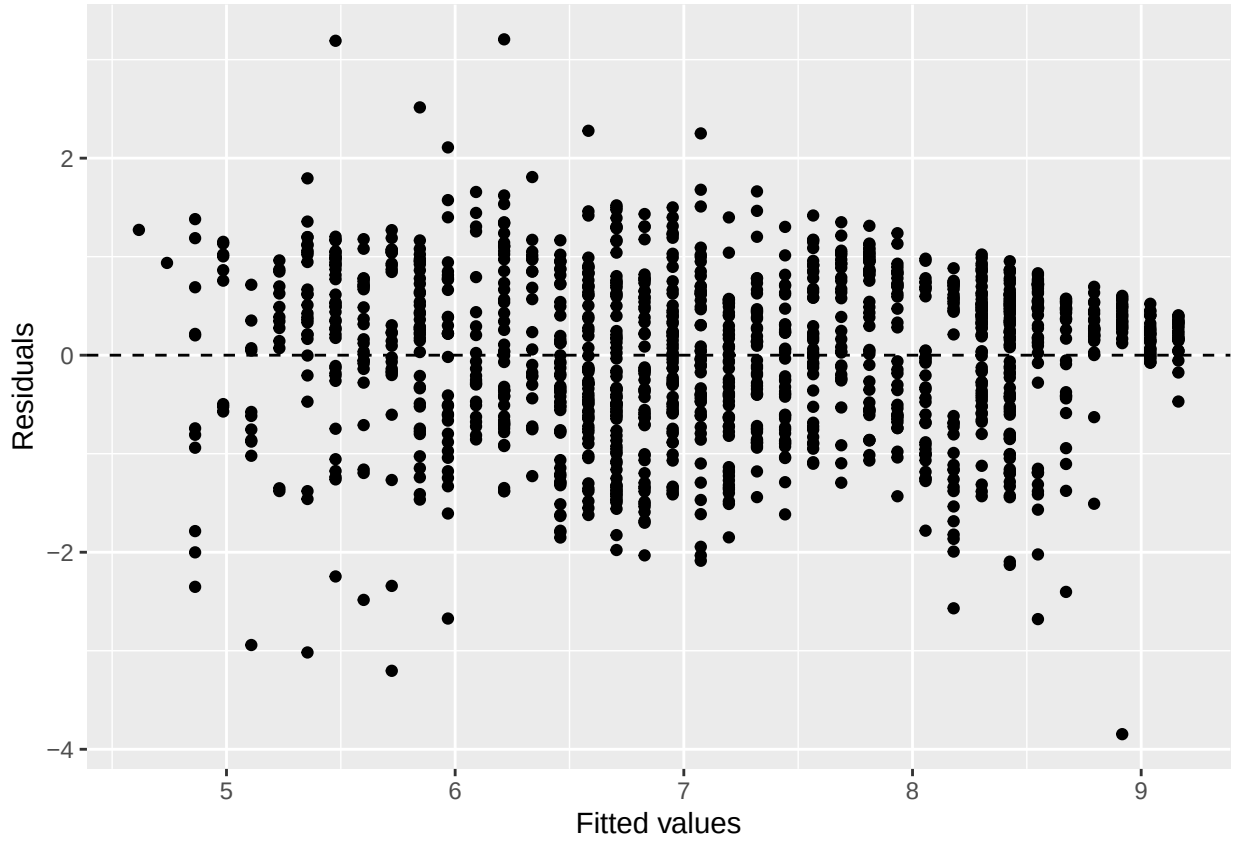
```
##        1
## 7.909663
```

Model diagnostics

To assess whether the linear model is reliable, we need to check for (1) linearity, (2) nearly normal residuals, and (3) constant variability.

Linearity: You already checked if the relationship between `pf_score` and 'pf_expression_control' is linear using a scatterplot. We should also verify this condition with a plot of the residuals vs. fitted (predicted) values.

```
ggplot(data = m1, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  xlab("Fitted values") +
  ylab("Residuals")
```

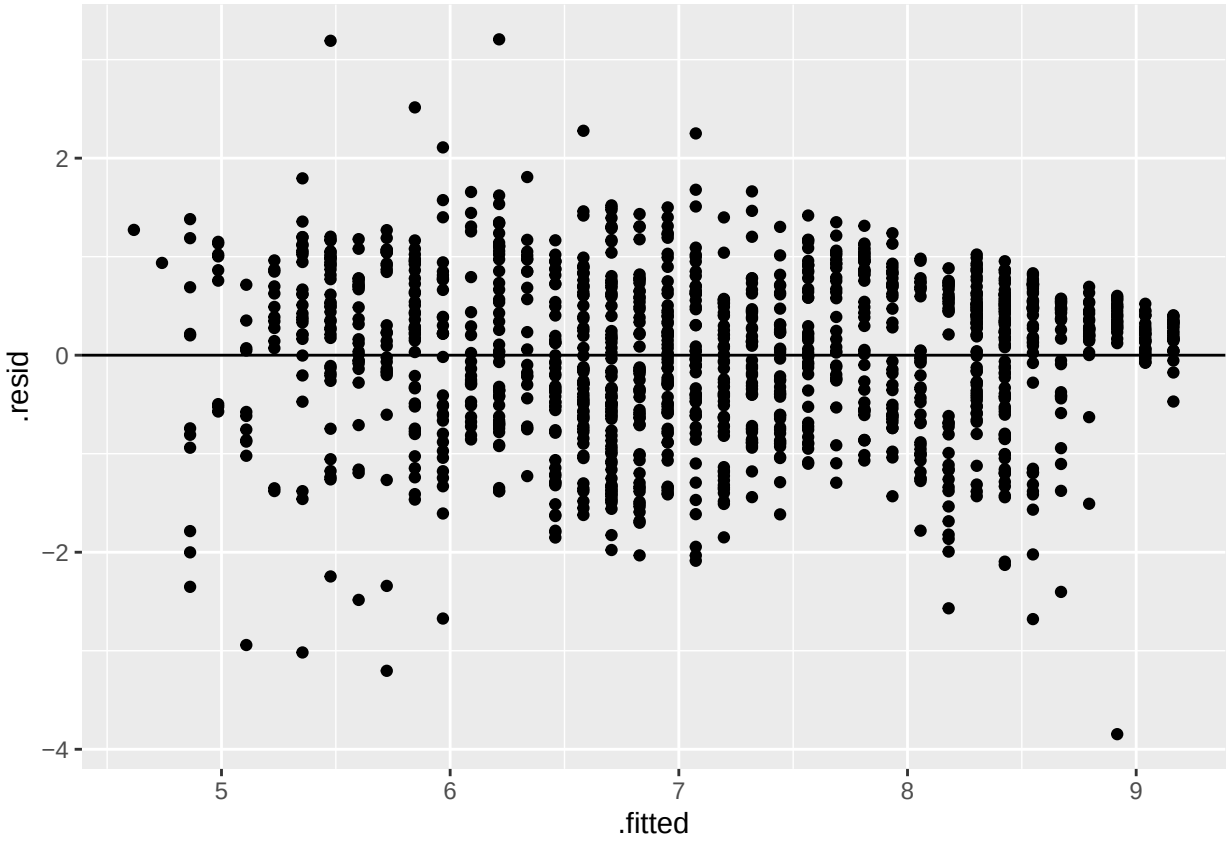


Notice here that `m1` can also serve as a data set because stored within it are the fitted values ($\hat{y}$) and the residuals. Also note that we're getting fancy with the code here. After creating the scatterplot on the first layer (first line of code), we overlay a horizontal dashed line at $y = 0$ (to help us check whether residuals are distributed around 0), and we also reanme the axis labels to be more informative.

7. Is there any apparent pattern in the residuals plot? What does this indicate about the linearity of the relationship between the two variables?

Insert your answer here

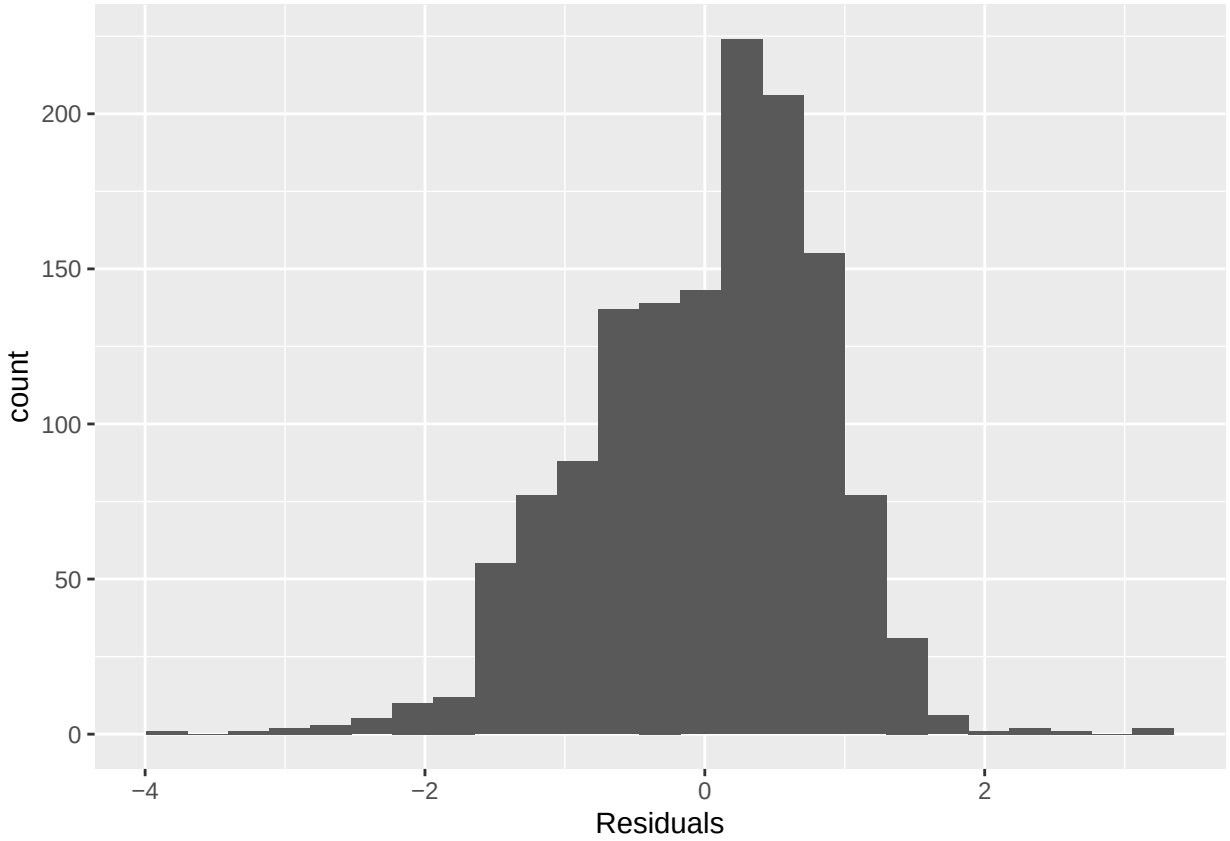
```
ggplot(data = m1, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0)
```



There is no strong curved or U-shaped pattern. The residuals are scattered around zero across the range of fitted values. This indicates that the relationship is reasonably linear and a linear model is appropriate. A few large negative residuals at the low end suggest some outliers, but no systematic bend.

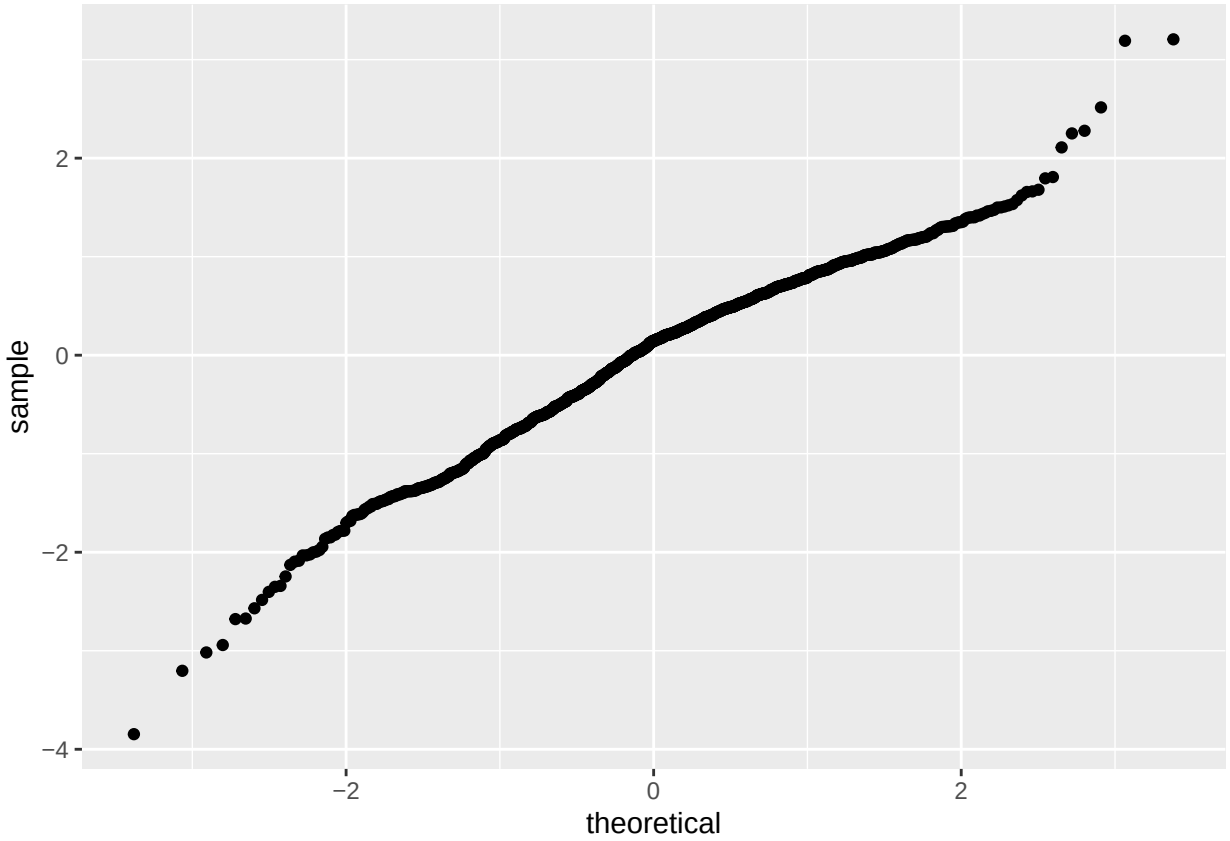
Nearly normal residuals: To check this condition, we can look at a histogram

```
ggplot(data = m1, aes(x = .resid)) +
  geom_histogram(bins = 25) +
  xlab("Residuals")
```



or a normal probability plot of the residuals.

```
ggplot(data = m1, aes(sample = .resid)) +
  stat_qq()
```



Note that the syntax for making a normal probability plot is a bit different than what you're used to seeing: we set `sample` equal to the residuals instead of `x`, and we set a statistical method `qq`, which stands for "quantile-quantile", another name commonly used for normal probability plots.

8. Based on the histogram and the normal probability plot, does the nearly normal residuals condition appear to be met?

Insert your answer here

Unimodal and roughly centered around zero, the histogram is negatively skewed, meaning that the right tail is longer than the left tail, extending from negative residual values up to about -3.8. Similarly, the normal probability plot follows the straight line in the middle but moves away from it on the lower side, which means it has a negative skew.

Constant variability:

9. Based on the residuals vs. fitted plot, does the constant variability condition appear to be met?

Insert your answer here

No. The spread of the residuals decreases with the increase in fitted values. For the lowest fitted values, the standard deviation of the residuals is 1.0, while for the highest ones, the standard deviation of the residuals is 0.6, which makes this graph a funnel-shaped one that narrows down from left to right. The condition of constant variance is fulfilled only approximately, which means that prediction is more accurate for high-freedom countries than for low-freedom countries.

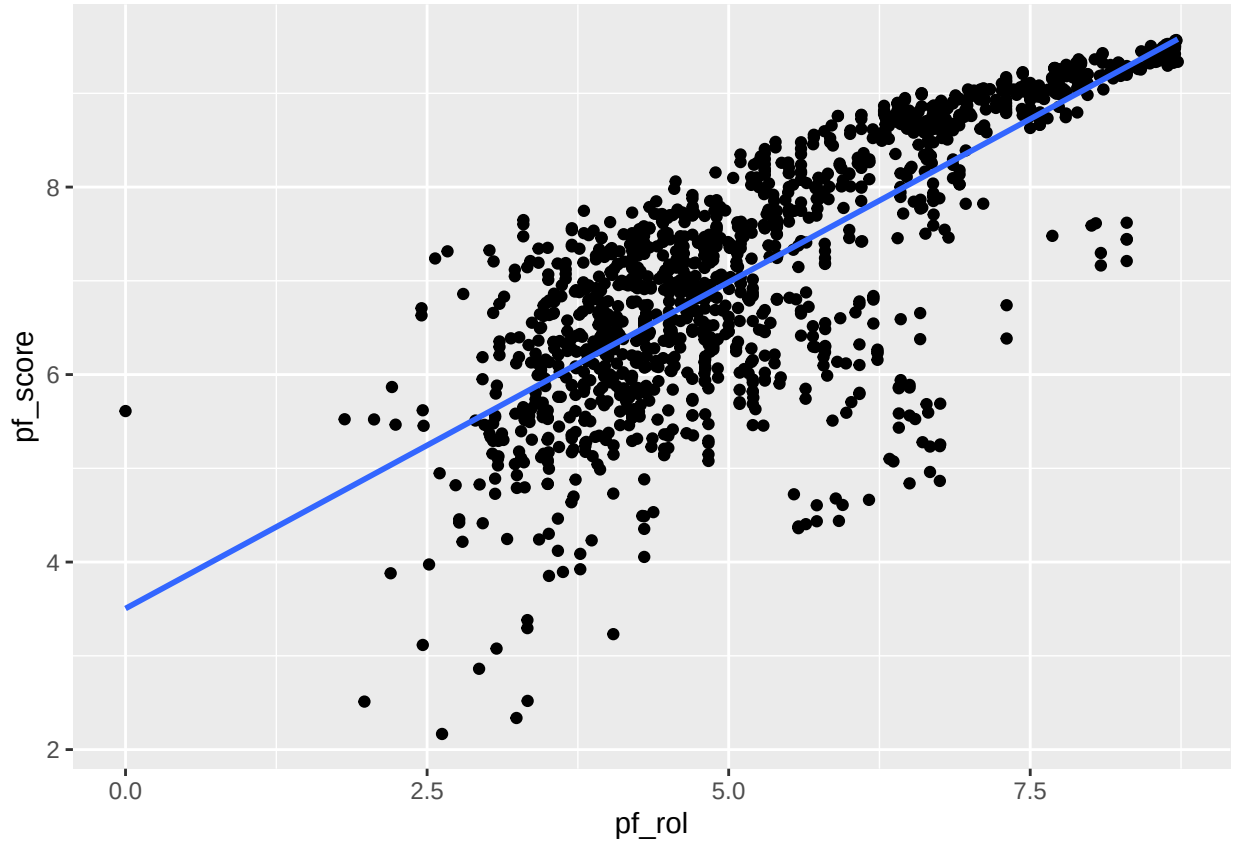
More Practice

- Choose another freedom variable and a variable you think would strongly correlate with it.. Produce a scatterplot of the two variables and fit a linear model. At a glance, does there seem to be a linear relationship?

Insert your answer here

I chose personal freedom score (pf_score) and the rule of law score (pf_rol), which I expected to be strongly related.

```
ggplot(data = hfi, aes(x = pf_rol, y = pf_score)) +
  geom_point() +
  stat_smooth(method = "lm", se = FALSE)
```




```
m3 <- lm(pf_score ~ pf_rol, data = hfi)
summary(m3)
```

```
##
## Call:
## lm(formula = pf_score ~ pf_rol, data = hfi)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.4241 -0.3700  0.1921  0.5717  2.1061
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.50458    0.08473   41.36   <2e-16 ***
```

```
## pf_rol       0.69622    0.01534   45.40   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8703 on 1376 degrees of freedom
##   (80 observations deleted due to missingness)
## Multiple R-squared:  0.5997, Adjusted R-squared:  0.5994
## F-statistic:  2061 on 1 and 1376 DF,  p-value: < 2.2e-16
```

At a glance there is a clear positive, roughly linear relationship. The fitted line is $\hat{pf_score} = 3.50 + 0.70 \times pf_rol$.

- How does this relationship compare to the relationship between `pf_expression_control` and `pf_score`? Use the R^2 values from the two model summaries to compare. Does your independent variable seem to predict your dependent one better? Why or why not?

Insert your answer here

For this model $R^2 = 0.5997$, compared with $R^2 = 0.6342$ for pf_expression_control and pf_score. Thus, pf_rol accounts for about 60 percent of the variance in pf_score, which is slightly lower than pf_expression_control (63.4%). Neither of them seems to be a superior predictor, as both explain almost all the variance. The possible explanation for why pf_rol performs a little worse is that rule of law is just one of the several dimensions of personal freedom, whereas media/expression control is closely related to the score.

- What's one freedom relationship you were most surprised about and why? Display the model diagnostics for the regression model analyzing this relationship.

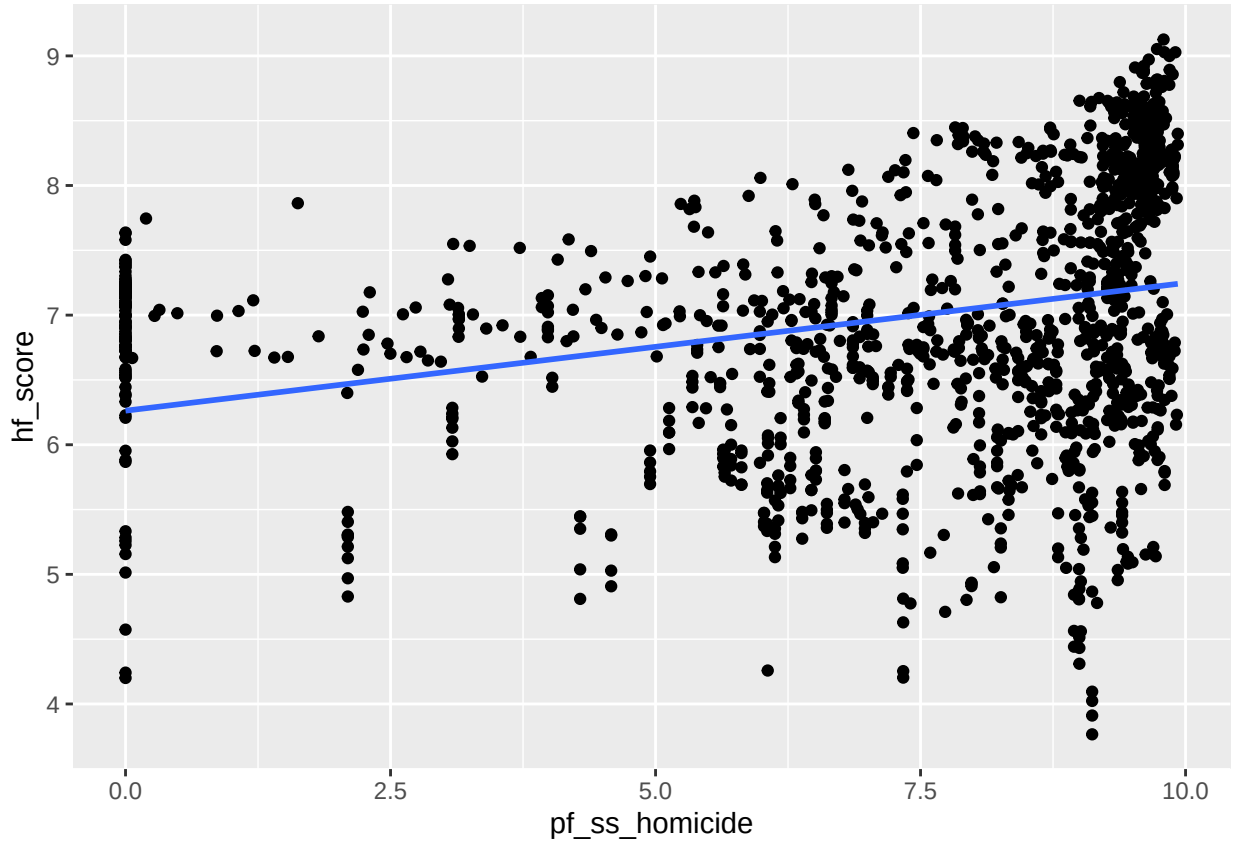
Insert your answer here

I was most surprised by how weakly personal safety and security from homicide (`pf_ss_homicide`) relates to the overall human freedom score (`hf_score`). I expected safer countries to be much freer, but the relationship is weak.

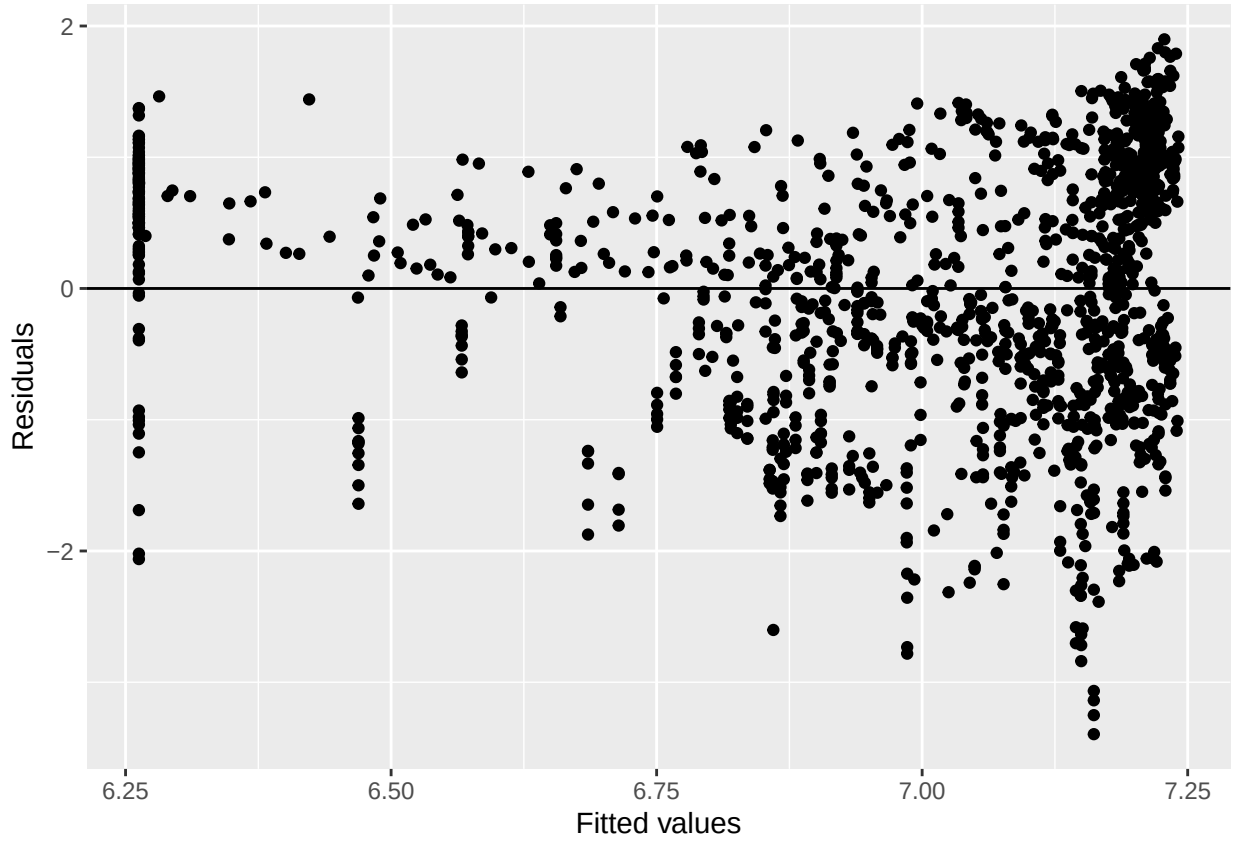
```
m4 <- lm(hf_score ~ pf_ss_homicide, data = hfi)
summary(m4)
```

```
##
## Call:
## lm(formula = hf_score ~ pf_ss_homicide, data = hfi)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.3958 -0.7063  0.0901  0.8563  1.8983
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.262567   0.074536   84.02   <2e-16 ***
## pf_ss_homicide 0.098594   0.009393   10.50   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9874 on 1376 degrees of freedom
##   (80 observations deleted due to missingness)
## Multiple R-squared:  0.07414,    Adjusted R-squared:  0.07347
## F-statistic: 110.2 on 1 and 1376 DF,  p-value: < 2.2e-16
```

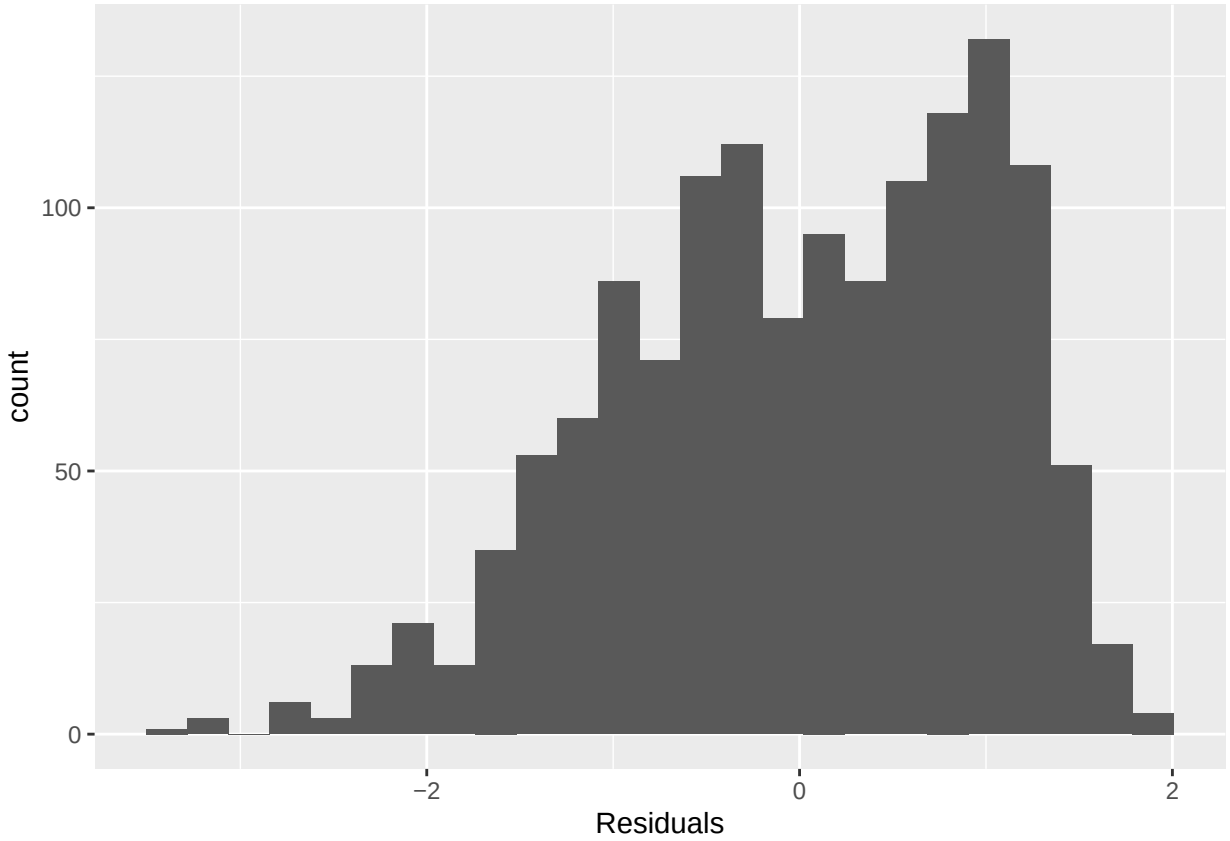
```
ggplot(data = hfi, aes(x = pf_ss_homicide, y = hf_score)) +
  geom_point() +
  stat_smooth(method = "lm", se = FALSE)
```



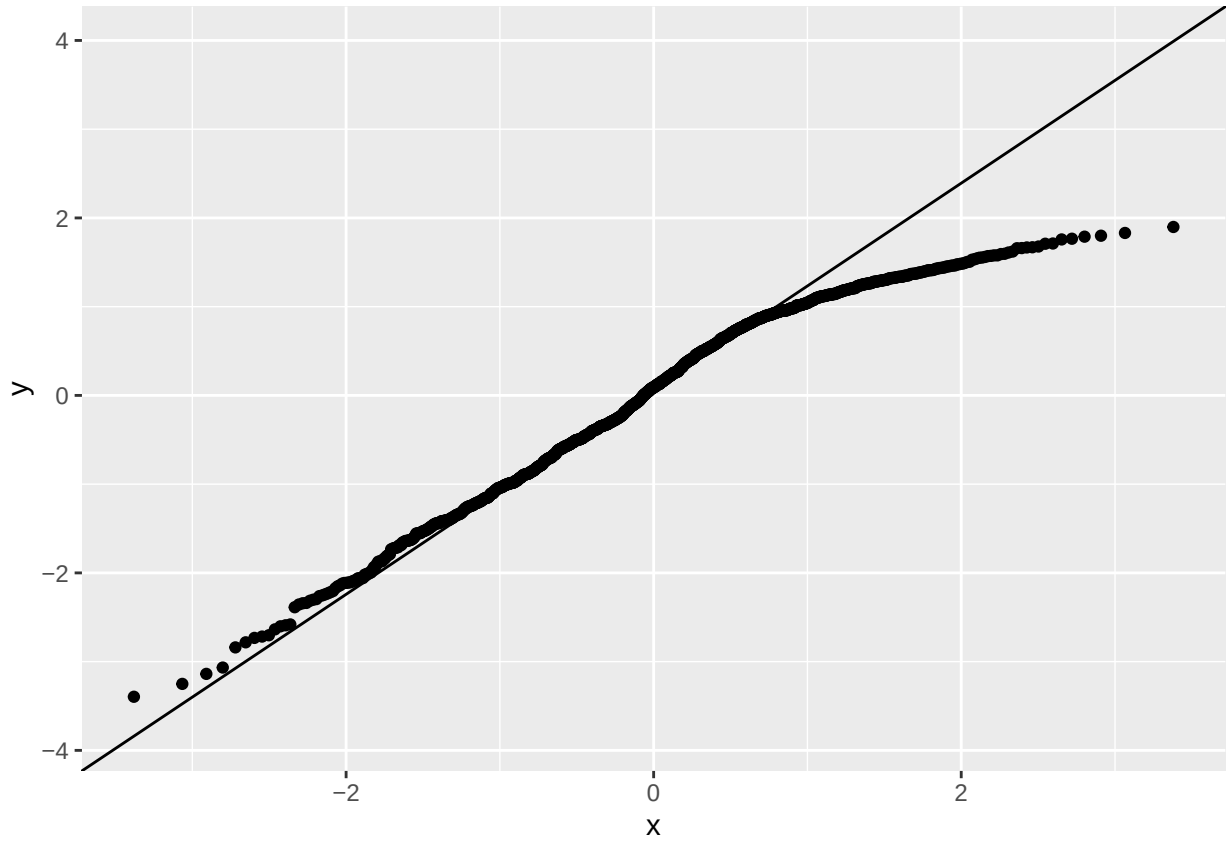

```
# Diagnostics
ggplot(data = m4, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0) +
  xlab("Fitted values") + ylab("Residuals")
```




```
ggplot(data = m4, aes(x = .resid)) +
  geom_histogram(bins = 25) +
  xlab("Residuals")
```




```
ggplot(data = m4, aes(sample = .resid)) +
  stat_qq() + stat_qq_line()
```



The slope is 0.10, while $R^2 = 0.074$. Homicide safety only accounts for 7% of the variance in total freedom. It can be seen from the diagnostics that the model does not fit well. The scatterplot has a wide fan pattern, where there are many observations close to large homicide values with diverse freedom scores, residuals are negatively skewed (for example, Syria 2016 has very low hf_scores but rather high homicide score). In hindsight, it can be noted that countries may have low homicide rates and at the same time repress freedom of speech or religion/market.
